

GOOGLE THEDAL

Organic Search Domination Engine

PROJECT DOCUMENT

LeadOS Platform | ABM Groups | BM TechX

Module Google Thedal v1.0	Platform LeadOS — app.abmggroups.org	Status Active Build
-------------------------------------	--	-------------------------------

1. Executive Summary

Google Thedal is the organic search intelligence module inside the LeadOS platform, built for BM TechX to deliver SEO-as-a-service to Tamil Nadu SMEs. While GMB Mafia controls Google Maps rankings, Thedal controls organic web search rankings — together forming a full Google domination stack.

Thedal = Search (Tamil)

When a business in Pondicherry or Tamil Nadu searches for their name, competitor, or service on Google — Thedal makes sure our clients appear first. We find them before they find us.

This is not a reporting tool. It is an automated SEO engine that tracks, alerts, generates content, audits, and delivers branded PDF reports — all without manual work.

STRATEGIC POSITION	
Problem it solves	Tamil Nadu SMEs spend on ads but have zero visibility into why competitors outrank them organically on Google
Solution	Automated daily rank tracking + GSC data + AI content + technical audit + monthly branded PDF report — all in one LeadOS module
Who pays for it	BM TechX charges clients ₹3,999–₹11,999/month. They get a full SEO team in a dashboard.
Revenue target	₹30,000 MRR in Month 1 from 5 clients → ₹85,000 Month 3 → ₹2.2 lakh Month 12
First client	Raahath Dental Care — generate free audit PDF, close on Growth plan ₹6,999/month

Competitive advantage	Only platform in Tamil Nadu that combines local language support (Tamil/Tanglish), WhatsApp rank alerts, and done-for-you content generation
------------------------------	--

2. What Thedal Does — Feature Map

#	Feature	What It Does	Automation
1	Keyword Rank Tracking	Daily rank check for every keyword via ValueSERP. Tracks position 1–100, rank change, SERP features (featured snippet, PAA, local pack)	n8n WF1 — 6 AM daily
2	GSC Intel	Pulls Google Search Console data: impressions, clicks, CTR, avg position, top queries, top pages, index coverage	n8n WF2 — Monday 5 AM
3	On-Page Audit	12-point technical SEO audit: H1, meta tags, HTTPS, page speed, image alt, schema, mobile-friendliness, robots.txt, sitemap, thin content, canonicals, broken links	n8n WF3 — Sunday 2 AM
4	Content Factory	AI-generated blog posts (English/Tamil/Tanglish), meta tag rewrites, topic calendar ideas, and JSON-LD schema markup via GPT-4o or Grok 3	On-demand in UI
5	Monthly PDF Report	Branded PDF with rank data, GSC metrics, audit score, top wins — auto-generated on 1st of every month for every client	n8n WF4 — 1st of month
6	Rank Drop Alert	WhatsApp alert to Kamar when any keyword drops > 5 positions or enters Top 10	n8n WF6 — after WF1
7	Client Onboard	First rank scan + first audit fires automatically when new client is added	n8n WF5 — webhook
8	SERP Radar	Live dashboard widget showing all clients' rank health at a glance	Real-time in UI
9	Gap Hunter	AI keyword opportunities — 15 high-value keywords the client should target	On-demand in UI
10	Schema Library	6 schema types (LocalBusiness, FAQPage, etc.) generated and ready to paste into client websites	On-demand in UI
11	Competitor Spy	Add rival domains, track their keywords, spot content gaps	Manual entry
12	Backlink Tracker	Log backlinks — source, anchor, DA, dofollow/nofollow, new/lost/toxic	Manual + Ahrefs Phase 2
13	Local Citations	Directory tracker — Justdial, Sulekha, Practo etc — track listed/pending/missing	Manual tracking
14	Local SEO Bridge	Correlation view: GMB Mafia Maps signals vs Thedal organic signals — Bundle clients only	Shared DB join

3. Technology Stack

FRONTEND	
Framework	Next.js 14 — App Router
Styling	Tailwind CSS — teal (#0D9488) primary color throughout
Charts	Recharts — rank trend lines, GSC line charts, audit score radial
Tables	TanStack Table v8 — keyword table with sort, filter, pagination
Animation	Framer Motion — SERP Radar pulse, rank change badges
Icons	Lucide React
State	Zustand — global client selection state
Auth	NextAuth.js JWT — shared session with LeadOS / GMB Mafia

BACKEND	
Runtime	Node.js 20 LTS + Express.js
ORM	Prisma 5 — type-safe queries to PostgreSQL 16
Database	PostgreSQL 16 — same DB as LeadOS/GMB Mafia (15 new thedal_* tables)
Cache	Redis 7 — rank results cached 20 hours, GSC data 1 hour
PDF	PDFKit — branded A4 monthly reports with rank table, GSC section
HTML Parser	Cheerio — homepage crawl for audit checks
Auth	JWT Bearer token — same middleware as LeadOS

EXTERNAL APIs	
ValueSERP	Keyword rank checking — same API key as GMB Mafia, one account covers both
Google Search Console API	Clicks, impressions, CTR, position, index coverage — OAuth2 per client
Google PageSpeed Insights	Core Web Vitals for audit — free API, no billing
OpenAI GPT-4o	Blog posts, meta rewrites, topic ideas, schema generation
Grok 3 (optional)	Swap in for Tamil/Tanglish content — change 2 lines in contentService.js

Meta WhatsApp Cloud API	Rank drop alerts — same integration as GMB Mafia
Ahrefs API	Backlinks — skip Phase 1, add Phase 2

AUTOMATION	
n8n (self-hosted)	6 automated workflows — same instance as GMB Mafia
WF1 — Daily Rank	Runs 6:00 AM IST daily — ValueSERP rank check for all active keywords
WF2 — GSC Sync	Runs Monday 5:00 AM IST — pulls GSC data for all connected clients
WF3 — Weekly Audit	Runs Sunday 2:00 AM IST — 12-point SEO audit for all clients
WF4 — Monthly Report	Runs 1st of month 7:00 AM IST — generates PDFs for all clients
WF5 — New Client	Webhook trigger — fires first scan + audit when new client added
WF6 — Rank Alerts	After WF1 — WhatsApp alert if rank drops > 5 or enters Top 10

4. Database Schema — 15 Tables

All tables use the thedal_ prefix. They live in the same PostgreSQL 16 database as LeadOS and GMB Mafia. No separate database required.

#	Table	Purpose	Key Fields
1	thedal_clients	Master client record — one row per website	domain, plan, gsc_connected, gmb_client_id
2	thedal_keywords	Keywords tracked per client	keyword, device, location, current_rank, best_rank
3	thedal_rank_history	Daily rank snapshots — timeseries data	keyword_id, checked_date, rank, serp_features
4	thedal_gsc_tokens	Google OAuth tokens per client (AES-256 encrypted)	access_token, refresh_token, expires_at
5	thedal_gsc_snapshots	Weekly GSC aggregate metrics	total_clicks, total_impressions, avg_ctr, avg_position
6	thedal_gsc_queries	Top search queries per client per sync	query, clicks, impressions, ctr, position
7	thedal_gsc_pages	Top pages per client per sync	page_url, clicks, impressions, ctr, position
8	thedal_gsc_coverage	Index coverage data	indexed_count, error_count, excluded_count

9	thedal_audits	Technical SEO audit runs	overall_score, pass_count, warn_count, fail_count
10	thedal_audit_items	Individual check results (12 per audit)	check_key, status, score, message, fix_guide
11	thedal_content	AI-generated content drafts	content_type, title, body, target_keyword, status
12	thedal_schema_library	Saved JSON-LD schema records	schema_type, schema_json, target_page, is_deployed
13	thedal_backlinks	Backlink records per client	source_domain, anchor_text, link_type, is_toxic
14	thedal_rivals	Competitor domains tracked	rival_domain, shared_keywords, content_gap_json
15	thedal_reports	Generated PDF report metadata	report_month, file_url, keywords_tracked, avg_rank

5. API Endpoints — Complete Map

Base: `/api/thedal` | Auth: `JWT Bearer token (user routes)` or `x-n8n-secret header (n8n routes)`

CLIENTS			
Method	Endpoint	Purpose	Auth
GET <code>/clients</code>	List all clients with KPIs	JWT	
POST <code>/clients</code>	Create client — triggers onboard webhook	JWT	
GET <code>/clients/:id</code>	Single client full data	JWT	
PUT <code>/clients/:id</code>	Update settings	JWT	
DELETE <code>/clients/:id</code>	Soft delete	JWT	

KEYWORDS			
Method	Endpoint	Purpose	Auth
GET <code>/keywords/:clientId</code>	All keywords with rank	JWT	
POST <code>/keywords/:clientId</code>	Add keyword	JWT	

DELETE /keywords/:clientId/:id	Remove keyword	JWT	
POST /keywords/bulk-check	Bulk rank scan — n8n WF1/WF5	n8n	
GET /keywords/:clientId/:id/history	Rank history for chart	JWT	
POST /keywords/:clientId/suggest	AI keyword suggestions	JWT	
GET /keywords/:clientId/cannibalization	Detect keyword cannibalization	JWT	

AUDIT

Method	Endpoint	Purpose	Auth
GET /audit/:clientId	Latest audit with 12 items	JWT	
GET /audit/:clientId/history	Audit score history	JWT	
POST /audit/run/:clientId	Trigger audit (async)	JWT	
POST /audit/score	Score raw data — n8n WF3	n8n	

CONTENT

Method	Endpoint	Purpose	Auth
GET /content/:clientId	Content calendar list	JWT	
POST /content/:clientId/generate-blog	AI blog post	JWT	
POST /content/:clientId/rewrite-meta	AI meta rewrite	JWT	
POST /content/:clientId/topic-ideas	8 topic ideas	JWT	
POST /content/:clientId/schema	JSON-LD schema	JWT	
PATCH /content/:clientId/:id/status	Publish/schedule	JWT	

REPORTS

Method	Endpoint	Purpose	Auth
GET /reports/:clientId	List reports	JWT	

POST /reports/generate/:clientId	Generate PDF — n8n WF4	n8n	
GET /reports/:clientId/:id/download	Download PDF	JWT	

GSC			
Method	Endpoint	Purpose	Auth
GET /gsc/connect/:clientId	Get Google OAuth URL	JWT	
GET /gsc/callback	OAuth redirect from Google	None	
GET /gsc/:clientId/summary	Clicks, impressions, CTR, position	JWT	
GET /gsc/:clientId/queries	Top 50 search queries	JWT	
GET /gsc/:clientId/pages	Top 20 pages	JWT	
GET /gsc/:clientId/coverage	Index coverage status	JWT	
POST /gsc/:clientId/sync	Manual GSC sync	JWT	

6. n8n Automation Workflows

WF	Name	Trigger	Cron (UTC)	Cron (IST)	What It Does
01	Daily Rank Check	Cron	30 0 * * *	6:00 AM daily	ValueSERP rank check all active keywords → update thedal_keywords + insert thedal_rank_history → trigger WF06
02	Weekly GSC Sync	Cron	30 23 * * 0	Monday 5:00 AM	Pull GSC data for all GSC-connected clients → upsert snapshots + queries + pages tables
03	Weekly Audit	Cron	30 20 * * 6	Sunday 2:00 AM	PageSpeed + HTML crawl + robots.txt for all clients → POST /api/thedal/audit/score → save 12 audit items

04	Monthly Reports	Cron	30 1 1 * *	1st 7:00 AM	POST /api/thedal/reports/generate for all clients → save PDF → WhatsApp Kamar when done
05	New Client Onboard	Webhook	POST /webhook/thedal/new-client	Instant	First rank scan + first audit → WhatsApp team: 'New client added, scans running'
06	Rank Drop Alert	Execute WF	After WF01	After 6 AM	Find drops > 5 positions + new Top 10 entries → WhatsApp Kamar with summary

n8n Credentials Required

1. LeadOS Postgres — same as GMB Mafia (skip if exists)
2. ValueSERP API — HTTP Header Auth | Header: Authorization | Value: Bearer YOUR_KEY
3. Meta WhatsApp — HTTP Header Auth (skip if exists from GMB Mafia)
4. Google GSC OAuth — Google OAuth2 Client ID + Secret

n8n Variables: N8N_INTERNAL_SECRET, GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, GOOGLE_PAGESPEED_API_KEY, WHATSAPP_PHONE_NUMBER_ID, META_WA_TOKEN, KAMAR_PHONE=9944309441

7. Frontend — 15 Screens

#	Screen	Route	What User Sees	Key Components
1	Thedal HQ	/thedal	Dashboard — all clients rank health, KPI cards, SERP Radar widget	ThedalHQ.jsx, SerpRadar.jsx
2	Clients	/thedal/clients	Client list — add new client modal, plan badge, keyword count	Clients.jsx
3	Keyword Map	/thedal/clients/:id/keywords	Keyword table with rank, delta, history chart sparkline, cannibalization tab	KeywordMap.jsx, Recharts
4	SERP Radar	/thedal/clients/:id/serp	All ranked keywords with rank badges, up/down indicators	SerpRadar.jsx
5	GSC Intel	/thedal/clients/:id/gsc	4 tabs: Overview metrics, Top Queries table, Top Pages table, Index Coverage	GscIntel.jsx, Recharts
6	Content Factory	/thedal/clients/:id/content	4 tabs: Blog Drafts generator, Meta Rewriter, Topic Ideas, Schema Library	ContentFactory.jsx

7	On-Page Audit	/thedal/clients/:id/audit	Score circle 0–100, 12-check accordion with fix guides, history chart	OnPageAudit.jsx
8	Schema Library	/thedal/clients/:id/schema	6 schema types, generate JSON-LD, copy-paste to website	SchemaLibrary.jsx
9	Backlink Tracker	/thedal/clients/:id/backlinks	Add/view backlinks — source, anchor, DA, type, new/lost/toxic	BacklinkTracker.jsx
10	Competitor Spy	/thedal/clients/:id/rivals	Add rival domains, view list, link to their sites	CompetitorSpy.jsx
11	Local Citations	/thedal/clients/:id/citations	10 major directories — track listed/pending/missing status	LocalCitations.jsx
12	Gap Hunter	/thedal/gap-hunter	AI keyword opportunity finder — select client, get 15 suggestions by difficulty	GapHunter.jsx
13	Reports	/thedal/reports	Monthly PDF list, generate button, download links	Reports.jsx
14	Settings	/thedal/settings	System status, workflow schedules, pricing plan reference	Settings.jsx
15	Layout	/thedal	Sidebar nav, teal color theme, active client sub-nav	ThedalLayout.jsx

8. Pricing Plans & Client Limits

Feature	Starter ₹3,999/mo	Growth ₹6,999/mo	Pro ₹11,999/mo	Bundle ₹10,999/mo
Keywords Tracked	5	15	30	15 (GMB+SEO)
Rank Check Frequency	Daily	Daily	Daily	Daily
GSC Integration	Basic	Full	Full	Full
Keyword History	30 days	60 days	90 days	60 days
Competitor Tracking	1 rival	3 rivals	5 rivals	3 rivals
Blog Drafts/month	2	4	8	4
Meta Rewrites/month	5	15	Unlimited	15
Technical Audit	Monthly	Bi-weekly	Weekly	Bi-weekly
Schema Library	3 types	All types	All types	All types
Local SEO Bridge	No	No	Yes	Yes (Bundle)
Gap Hunter	No	Yes	Yes	Yes

Branded PDF Report	Monthly	Monthly	Monthly	Monthly (both modules)
WhatsApp Rank Alerts	Weekly digest	Daily digest	Real-time	Daily digest
Support	Email	WhatsApp	WhatsApp Priority	WhatsApp
Includes GMB Mafia	No	No	No	Yes (saves ₹3,000)

9. Revenue Model & Projections

REVENUE TARGETS	
Month 1 — 5 clients	₹30,000+ MRR Raahath Dental (Growth) + 4 existing BM TechX clients
Month 3 — 15 clients	₹85,000 MRR 5 Starter + 7 Growth + 3 Pro
Month 6 — 22 clients	₹1,40,000 MRR Scaling through referrals + campus network
Month 12 — 30+ clients	₹2,20,000 MRR 20 GMB Mafia + 10 Thedal only + bundle clients
Year 1 Total Revenue	₹18–22 lakh/year from Thedal alone (stacks with GMB Mafia revenue)

IMMEDIATE CONVERSION TARGETS (Month 1)	
Raahath Dental Care	Growth plan — ₹6,999/mo Hook: free audit PDF showing 52/100 score
Ramya Agencies	Starter plan — ₹3,999/mo Already a BM TechX client
Vajra Doors	Growth plan — ₹6,999/mo Already a BM TechX client
CMI Pondicherry	Pro plan — ₹11,999/mo High-value local institution
Mishi Sambrani	Starter plan — ₹3,999/mo Low-effort conversion
Total Day 1 MRR	₹30,999/month from existing relationships alone

Sales Hook Script for Raahath Dental
Step 1: Run audit from Thedal UI → generates PDF with score (likely 40–65/100)
Step 2: WhatsApp PDF to Dr. Raahath: 'We ran a free SEO audit — your score is X/100. Competitors are at 75+'
Step 3: Call: 'Doctor, right now when someone searches dentist pondicherry — you're on page 2. They call your competitor.'
Step 4: 'Growth plan ₹6,999/month. We fix all 12 issues in 30 days. First month — no improvement = no charge for month 2.'
Close rate with this script: 80%+ (Kamar closes personally for first 5 clients)

10. External API Dependencies

API	Used For	Setup Time	Cost	Shared with GMB Mafia?
ValueSERP	Keyword rank checking (positions 1–100 in organic search)	5 min — signup only	~\$12/mo Starter plan	YES — same API key
Google PageSpeed	Core Web Vitals in audit (mobile score, LCP)	10 min — Cloud Console	Free — no charge	No — new key
Google Search Console	GSC clicks, impressions, CTR, queries — OAuth per client	20 min — Cloud setup	Free	No — new OAuth
OpenAI / GPT-4o	Blog generation, meta rewrite, topic ideas, schema	5 min — platform.openai.com	~\$5–20/mo usage	No — new key
Meta WhatsApp	Rank drop alerts, monthly report notification to Kamar	Already set up	Already paying	YES — same token
Ahrefs API	Backlink data (Phase 2 only — skip now)	Phase 2	\$99/mo (skip Phase 1)	No

ValueSERP — One Key for Both GMB Mafia and Thedal

GMB Mafia uses ValueSERP to check if client is in Google Maps 3-pack (local pack position #1, #2, #3)
Thedal uses ValueSERP to check if client is in organic search results (position 1–100 in web results)
Same API key. Different search parameters. Cost at Month 1 (5 clients): ~\$2–5/month total
Buy: trajectdata.com/serp/value-serp-api — Start Free Trial (100 free searches for dev testing)

11. Deployment Overview

INFRASTRUCTURE	
Server	Existing VPS (Ubuntu 24.04) — same server as LeadOS and GMB Mafia
Domain	app.abmggroups.org — Thedal routes as /thedal/* and /api/thedal/*
Database	PostgreSQL 16 — existing LeadOS DB + 15 new thedal_* tables
Node.js	v20 LTS — already installed
Redis	Already running — Thedal adds thedal: prefix to avoid GMB Mafia key collision
n8n	Already running — 6 new Thedal workflows added to same instance

Nginx	Already configured — add /api/thedal/ proxy block
SSL	Let's Encrypt already active — no new cert needed
Reports directory	Create: /var/leados/reports/thedal/

Step	Task	Time	How
1	VPS prerequisites check	15 min	Verify Node 20, Redis, PM2, n8n all running
2	Add .env variables	10 min	Append 10 new variables to existing .env
3	Prisma migration	10 min	npx prisma db push — creates 15 thedal_* tables
4	npm install	5 min	npm install axios redis cheerio googleapis openai pdfkit
5	Copy backend files	15 min	5 services + 7 routes + 1 middleware into correct folders
6	Mount router in server.js	2 min	Add 2 lines: import + app.use('/api/thedal', thedalRouter)
7	Nginx update	10 min	Add /api/thedal/ proxy block (may already work)
8	PM2 restart + test	10 min	pm2 restart leados-backend curl /api/thedal/health
9	Import n8n workflows	30 min	Import 6 JSON files in correct order, set credentials
10	Deploy frontend	30 min	Copy 15 JSX screens, add routes to App.jsx, npm run build
11	Smoke test	30 min	Backend + n8n + frontend checklist — 25 items
TOTAL		~2.5 hours	For an experienced full-stack developer

12. File Delivery Structure

All 57 files are delivered in the GOOGLE-THEDAL-DELIVERY.zip package, organised into folders ready to copy into the LeadOS codebase.

```
GOOGLE-THEDAL-DELIVERY/
├── README.md                ← Developer reads this first — step-by-step guide
├── 00_READ_FIRST/
│   └── THEDAL_DEV_HANDOFF.docx ← Complete step-by-step with all commands
├── 01_database/
│   ├── schema_thedal_addition.prisma ← Append to prisma/schema.prisma
│   ├── seed_thedal_clients.sql       ← Run once — 7 ABM brands seeded
│   └── prisma.js                     ← lib/prisma.js singleton (if missing)
├── 02_backend/
│   └── services/
```

13. First Client — Raahath Dental Care

ONBOARDING SEQUENCE (Day of Go-Live)

Step 1 — Add client Thedal UI → Clients → + New Client | Business: Raahath Dental Care | Domain: raahathdentalcare.in | Type: Dental Clinic | Plan: Growth

Step 2 — Add 5 keywords	dentist pondicherry dental clinic near me pondicherry teeth whitening pondicherry best dentist pondy dental implants pondicherry
Step 3 — Run rank check	Click Run Rank Check → wait 60 seconds → positions appear for all 5 keywords
Step 4 — Run audit	On-Page Audit tab → Run Audit → wait 3 minutes → score circle appears (likely 40–65/100)
Step 5 — Connect GSC	GSC Intel tab → Connect Google Search Console → Dr. Raahath logs into his Google account → authorises
Step 6 — Generate PDF	Reports tab → Generate Report → download branded PDF with rank data + audit score
Step 7 — WhatsApp PDF	Send PDF to Dr. Raahath: 'Doctor, we ran a free SEO audit — your score is X/100. Competitors at 75+'
Step 8 — Close the deal	Call: 'Growth plan ₹6,999/month. We fix all issues in 30 days. No results = no charge month 2.'

14. Common Errors & Fixes

Error Message	Root Cause	Fix
Cannot find module 'cheerio'	Package not installed	npm install cheerio in backend folder
TOKEN_ENCRYPTION_KEY must be 32 bytes	Wrong .env value length	openssl rand -hex 32 → must be exactly 64 hex chars
Prisma model ThedalClient not found	Client not generated	npx prisma generate
thedal_clients table doesn't exist	Migration not run	npx prisma db push
n8n route returns 401	Secret mismatch	N8N_INTERNAL_SECRET in .env must match n8n Variables
GSC OAuth redirect_uri mismatch	Wrong redirect URI	Must be: https://app.abmggroups.org/api/thedal/gsc/callback
WF01 ValueSERP 401 error	Wrong or expired API key	Check VALUESERP_API_KEY in .env and n8n Variables
/thedal returns 404	Routes not added	Add thedal routes to App.jsx — see thedal_routes.jsx
PDF not generating	Permissions issue	mkdir -p /var/leados/reports/thedal && chown -R \$USER .
GSC shows 0 data	Client not connected	Client must complete OAuth via GSC Intel tab

15. Key Contacts & Reference

CONTACTS	
Kamar (Founder, ABM Groups)	WhatsApp: +91 99442 88271 admin@abmggroups.org
Satish (Ops / Dev delivery)	Execution and delivery coordination
Dev questions	WhatsApp Kamar before 10 PM — responds within 1 hour
VPS / Server access	Kamar provides SSH credentials to dev directly
Google Cloud account	Kamar shares access credentials for Cloud Console setup

KEY URLS & CREDENTIALS	
Live platform	https://app.abmggroups.org
n8n dashboard	https://app.abmggroups.org:5678
Health check endpoint	https://app.abmggroups.org/api/thedal/health
Thedal module (after deploy)	https://app.abmggroups.org/thedal
ValueSERP signup	trajectdata.com/serp/value-serp-api
Google Cloud Console	console.cloud.google.com
OpenAI API keys	platform.openai.com/api-keys
Reports storage path	/var/leadoss/reports/thedal/
n8n webhook path	https://app.abmggroups.org/n8n/webhook/thedal/new-client

Build it. Ship it. Dominate Tamil Nadu Google search.

ABM Groups | BM TechX | LeadOS | app.abmggroups.org